Math 124F **Midterm 2** **Autumn 2021**

Your Name

Your Signature

Student ID #

- Turn off all cell phones, pagers, radios, mp3 players, and other similar devices.
- Please write your name at the top of every page.
- This exam is closed book. You may use one $8.5'' \times 11''$ sheet of handwritten notes (both sides OK). Do not share notes. No photocopied materials are allowed.
- You can use only a Texas Instruments TI-30X IIS calculator.
- In order to receive credit, you must **show all of your work**. If you do not indicate the way in which you solved a problem, you may get little or no credit for it, even if your answer is correct.
- Place $\boxed{\text{a box around your answer}}$ to each question.
- The pages have problems on **both** sides.
- If you need more room, use the blank last page and indicate that you have done so.
- Raise your hand if you have a question.
- This exam has 5 pages, plus this cover sheet. Please make sure that your exam is complete.

1. (12 total points) Compute the derivatives of the following functions. Do not simplify your answers.

   (a) (4 points) $f(x) = \sqrt{\cos^2 x + 5x^7}$

   (b) (4 points) $g(t) = \tan^{-1}\left(\dfrac{5t+3}{t^2+4}\right)$

   (c) (4 points) $h(x) = \sin\left(2 + \sin\sqrt{1+x^3}\right)$

2. (12 total points) An object is moving along an ellipse. Its location is given by the parametric equations

$$x(t) = 1 + 2\cos t \qquad y(t) = 2 + 4\sin t$$

In this problem we take $0 \le t \le 2\pi$.

(a) (6 points) Find a formula that gives the slope of the tangent line to the path at time t as a function of t.

(b) (6 points) Find the equation of the tangent line at $t = \frac{\pi}{3}$.

3. (8 points) Compute the equation of the tangent line to the curve $y = x^{\cos(\pi x)}$ at the point where $x = 2$. Use **exact values** only. Do not give decimal approximations.

4. (8 points) If $x^3 + xy + y^2 = 8$, find the value of y'' at all points where $x = 2$.

5. (10 points) Find all the points (x,y) on the curve

$$x = t^2 + 8t, \quad y = 2t + 3\ln t$$

where the tangent line is parallel to the line $x - 2y = 7$.

This page is for extra work.